

Algorithm Analysis And Data Structures

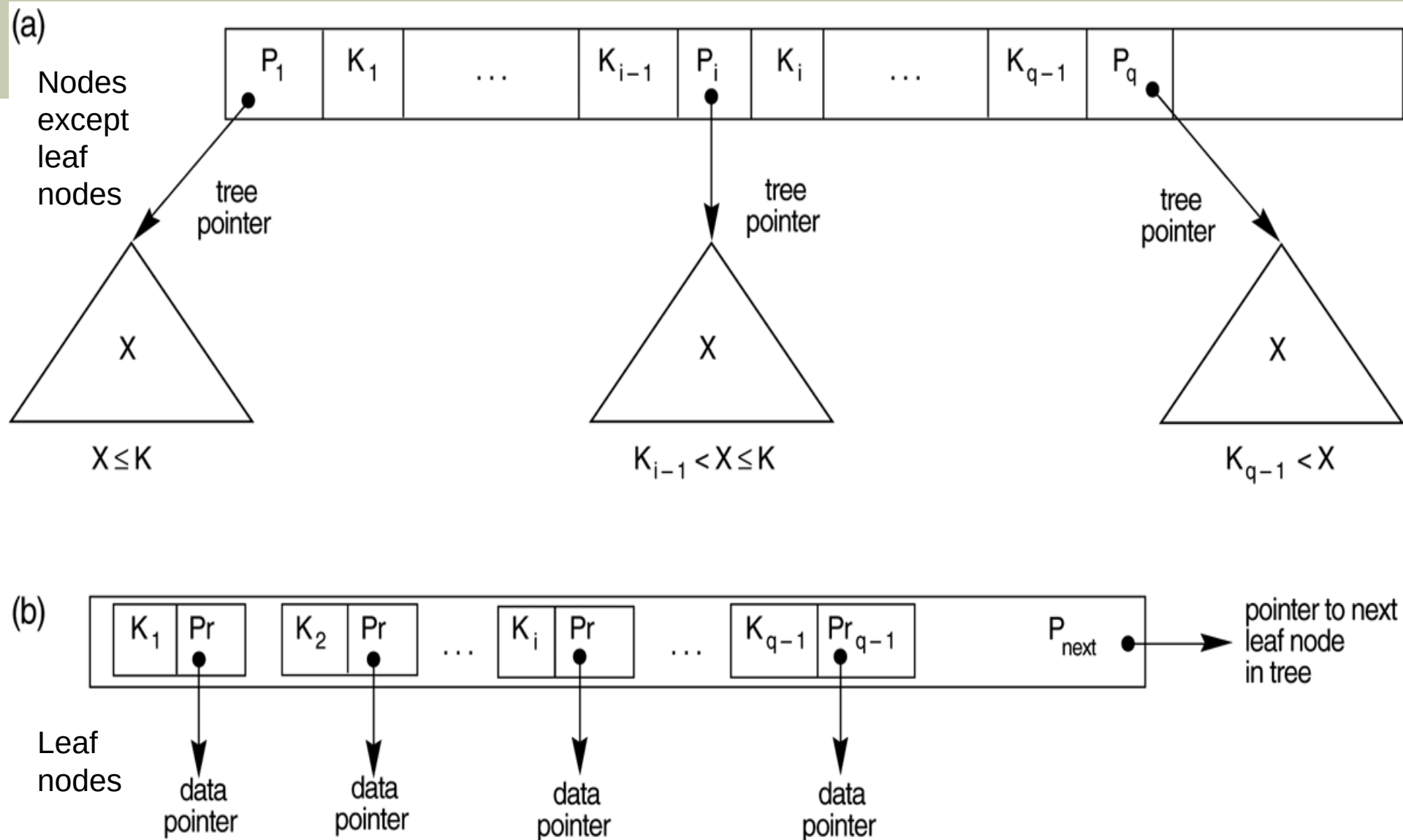
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What is a B+Tree?

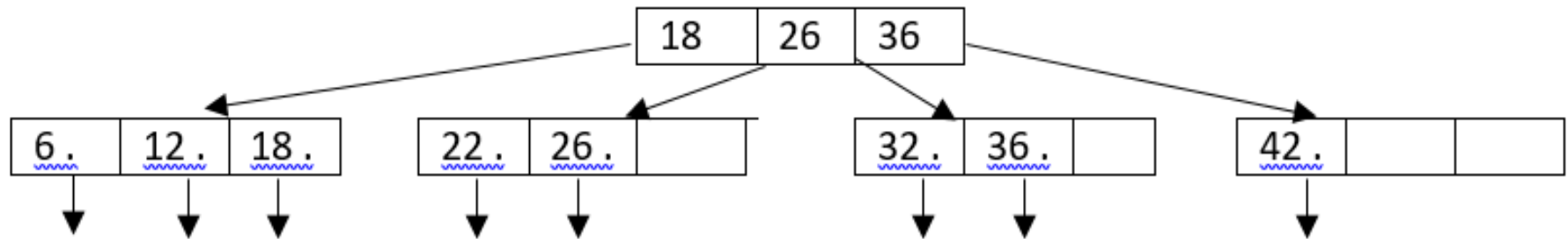
- B+ tree of order m (or branching factor m), where $m > 2$, is either an empty tree or a multiway search tree with the following properties:
 - The root is either a leaf or it has at least two non-empty subtrees and at most m non-empty subtrees.
 - Each non-leaf node, other than the root, has at least $\lceil m/2 \rceil$ non-empty subtrees and at most m non-empty subtrees. (Note: $\lceil x \rceil$ is the lowest integer $> x$).
 - The number of keys in each non-leaf node is one less than the number of non-empty subtrees for that node. Keys are placed according to a and b (which are shown in the following page) in node structures.
 - All leaf nodes are at the same level; that is the tree is perfectly balanced.

The node structure of a B+ tree

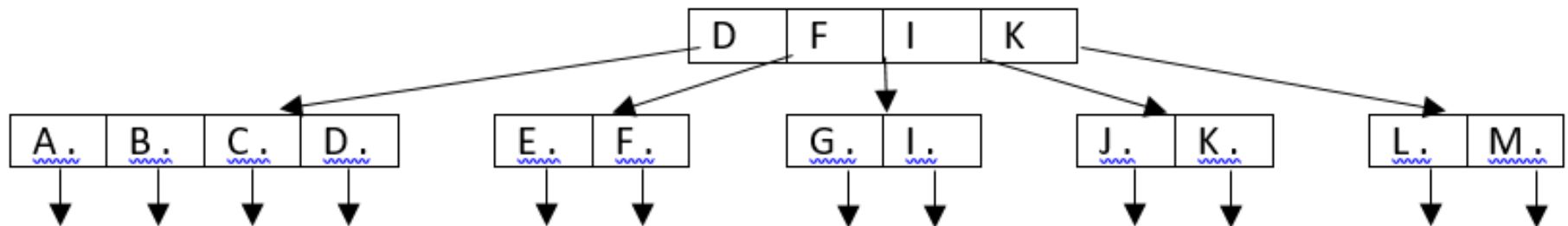


B+ Tree Examples

Example: B+ tree of order(way) 4



Example: B+ tree of order(way) 5



Deletion in B+Tree

- Like insertion, deletion must be on a leaf node. If the key to be deleted is not in a leaf, swap it with either its successor or predecessor (each will be in a leaf).
- The successor of a key k is the smallest key greater than k .
- The predecessor of a key k is the largest key smaller than k .
- IN B+TREE THE SUCCESSOR AND PREDECESSOR, IF ANY, OF ANY KEY IS IN A LEAF NODE

Deletion in B+Tree

- **UNDERFLOW CONDITION**
- A non-root node of a B+ tree of order m underflows if, after a key deletion, it contains $\lceil m / 2 \rceil - 2$ keys
- The root node does not underflow. If it contains only one key and this key is deleted, the tree becomes empty.

■ Deletion algorithm:

If a node underflows, rotate the appropriate key from the adjacent right- or left-sibling if the sibling contains at least $\lceil m / 2 \rceil$ keys; otherwise perform a merging.

⇒ A key rotation must always be attempted before a merging

■ There are five deletion cases:

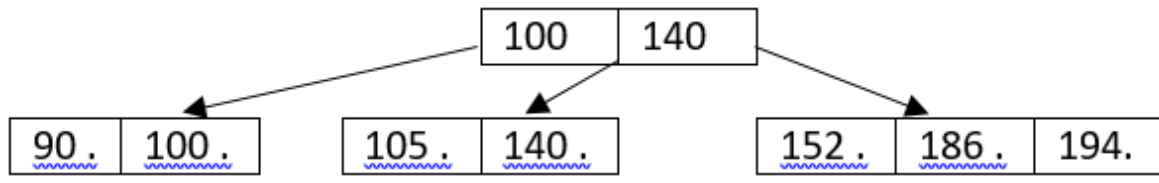
- 1) The leaf does not underflow.
- 2) The leaf underflows and the adjacent right sibling has at least $\lceil m / 2 \rceil$ keys: **perform a left key-rotation**
- 3) The leaf underflows and the adjacent left sibling has at least $\lceil m / 2 \rceil$ keys: **perform a right key-rotation**
- 4) The leaf underflows and each of the adjacent right sibling and the adjacent left sibling has at least $\lceil m / 2 \rceil$ keys: **perform either a left or a right key-rotation**
- 5) The leaf underflows and each adjacent sibling has $\lceil m / 2 \rceil - 1$ keys: **perform a merging**

Deletion in B+Tree

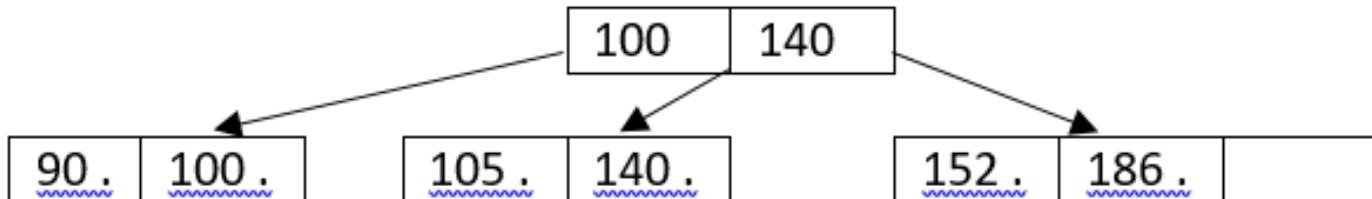
Case1: The leaf does not underflow.

Example:

B+tree of order 5



Delete 194

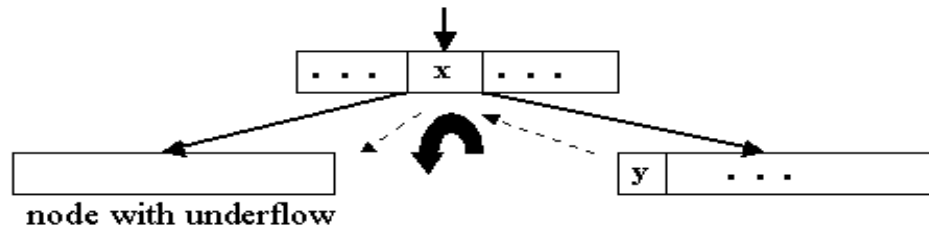


Deletion in B+Tree

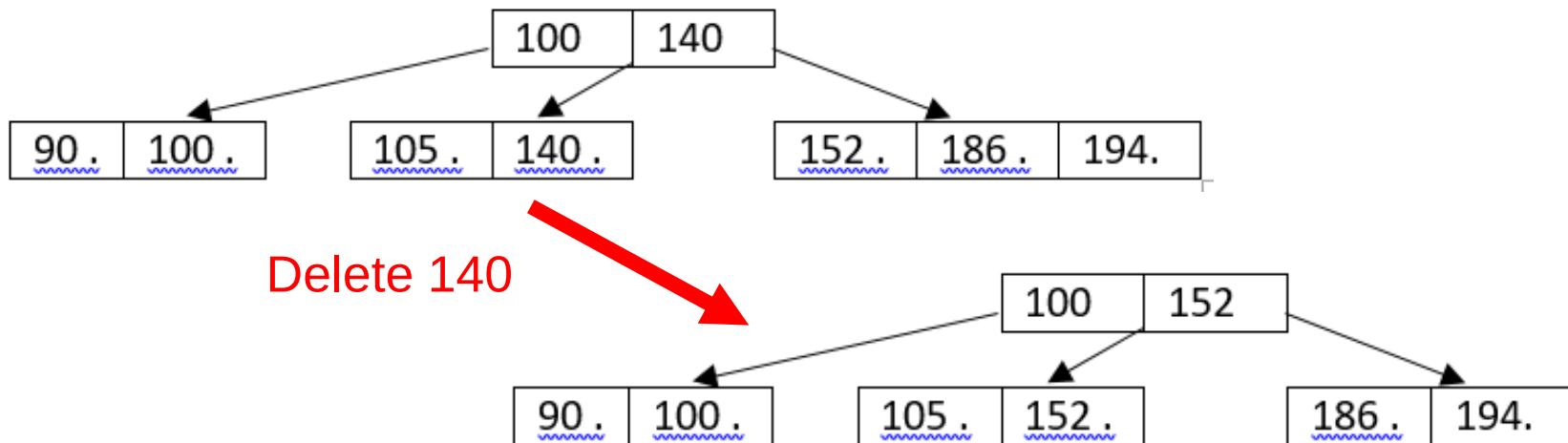
Case2: The leaf underflows and the adjacent right sibling has at least $\lceil m / 2 \rceil$ keys.

Perform a left key-rotation:

1. Move the parent key x that separates the siblings to the node with underflow
2. Move y , the minimum key in the right sibling, to where the key x was
3. Make the old left subtree of y to be the new right subtree of x .



Example: B-tree of order 5

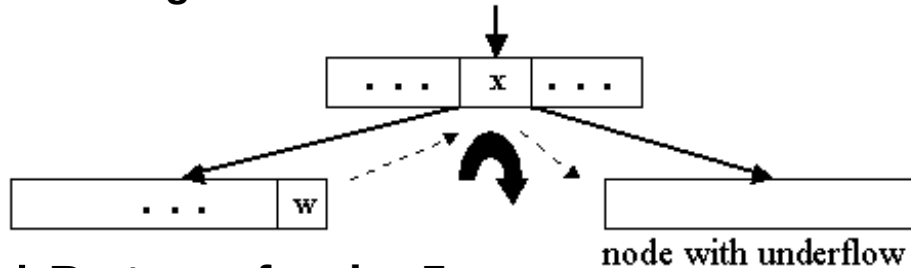


Deletion in B+Tree

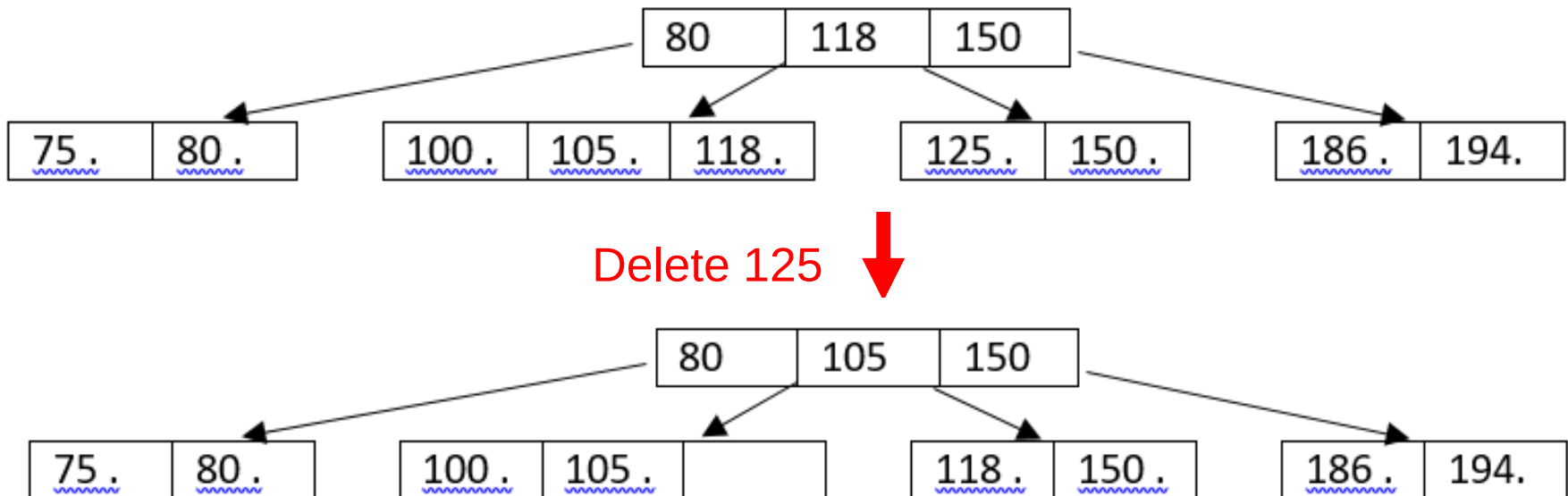
Case 3: The leaf underflows and the adjacent left sibling has at least $\lceil m / 2 \rceil$ keys.

Perform a right key-rotation:

1. Move the parent key x that separates the siblings to the node with underflow
2. Move w , the maximum key in the left sibling, to where the key x was
3. Make the old right subtree of w to be the new left subtree of x



Example: B+ tree of order 5

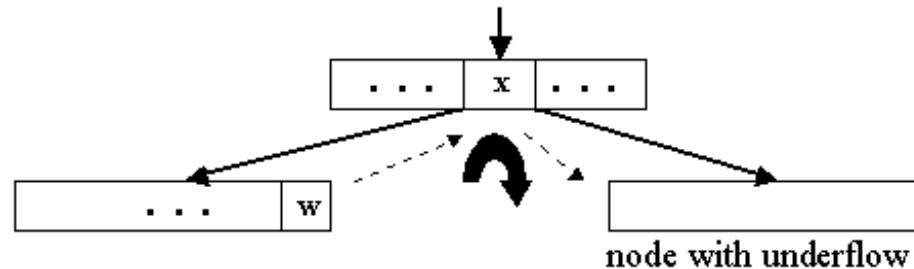


Deletion in B+Tree

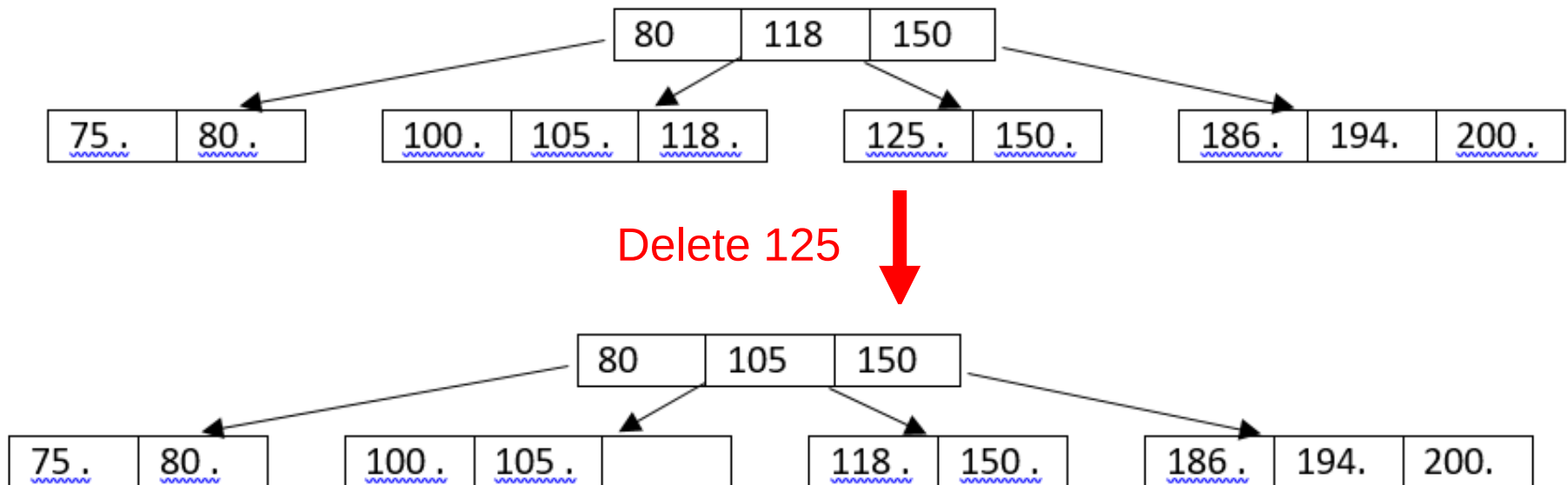
Case 4: The leaf underflows and each of the adjacent right sibling and the adjacent left sibling has at least $\lceil m / 2 \rceil$ keys:

perform either a left or a right key-rotation

In order to apply the same rotation for this course, we will prefer the right key rotation

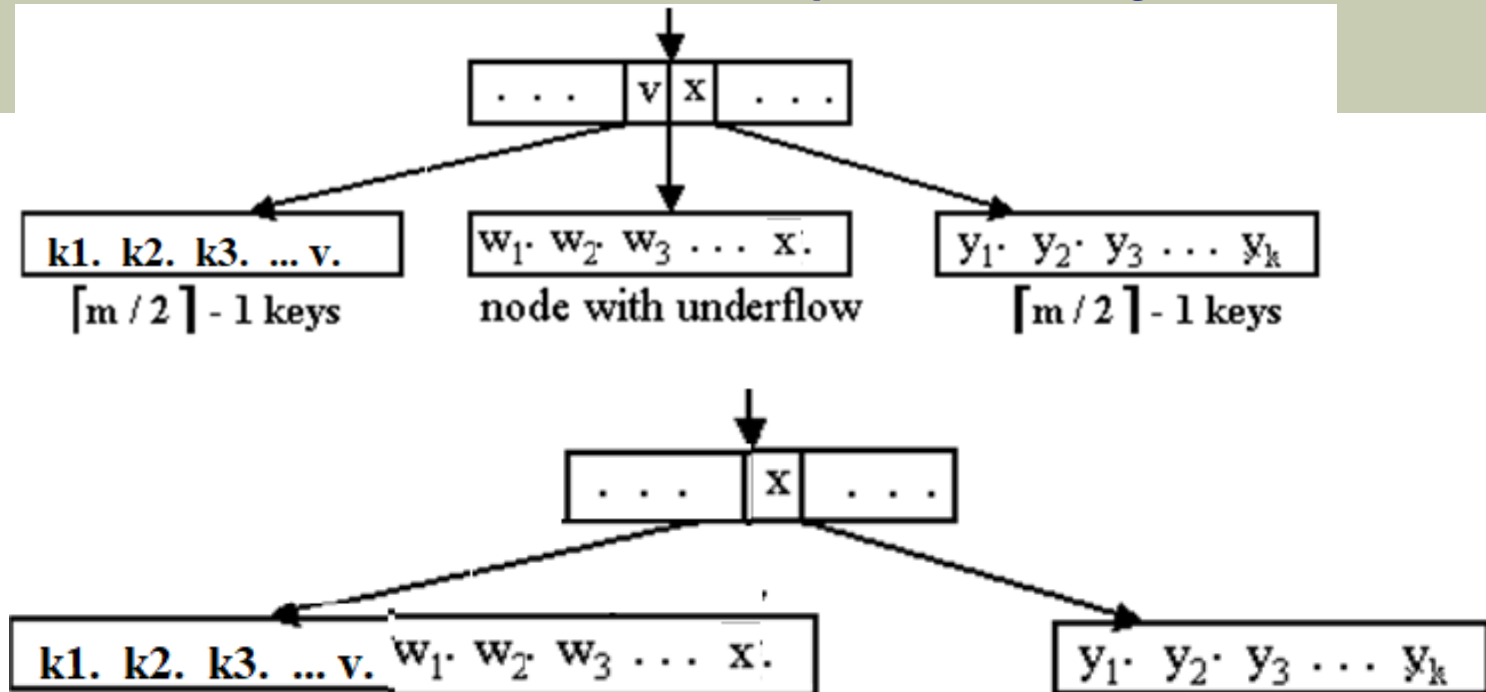


Example: B+ tree of order 5



Deletion in B-Tree

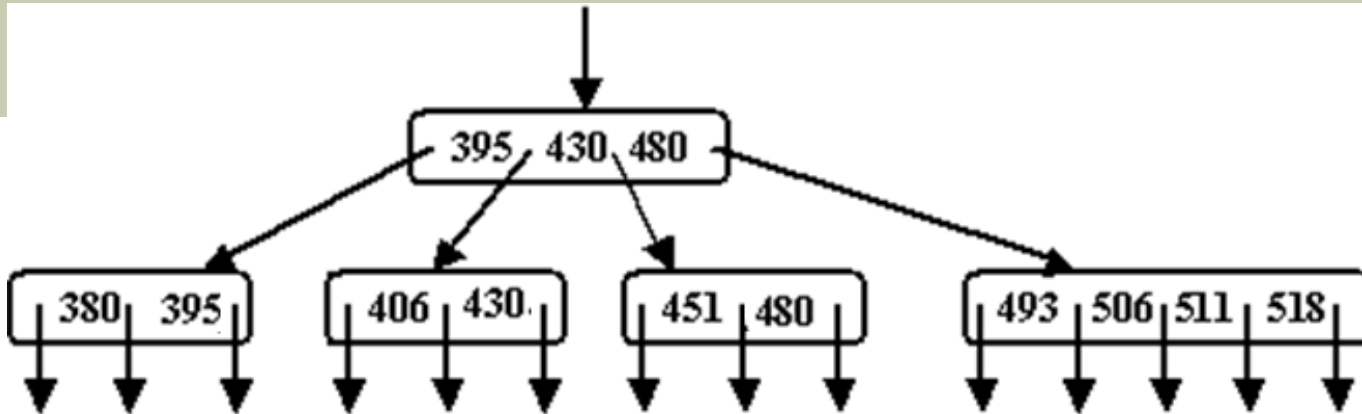
Case 5: The leaf underflows and each adjacent sibling has $\lceil m/2 \rceil - 1$ keys.



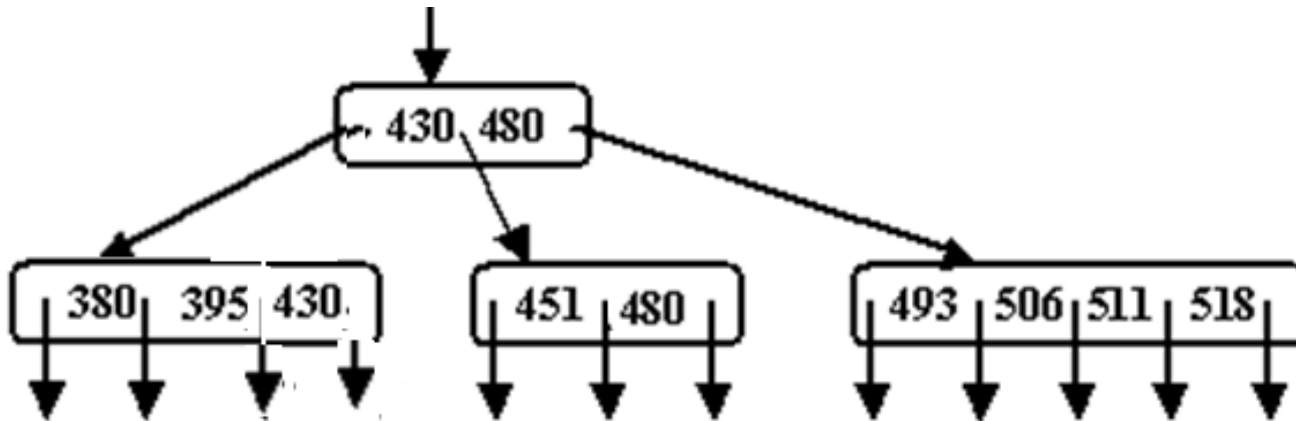
If the parent of the merged node underflows, the merging process propagates upward. In the limit, a root with one key is deleted and the height decreases by one.

Note: The merging could also be done by using the left sibling instead of the right sibling for this course

Case 5 - Example: B+ tree of order 5



Delete 406 ↓



The parent of the merged node does not underflow. The merging process does not propagate upward.

Deletion in B+Tree

Example: B+ tree of order 5

